Paleomagnetism of the Sutton Lake Proterozoic inlier, Ontario, Canada

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Thirty-eight cores were collected from eight sites in the Sutton Lake Proterozoic inlier, which is thought to be an extension of the Circum-Ungava Belt. Six sites from a 100 m thick diabase sill yielded essentially single-component magnetization averaging $D = 35°$, $I = -54°$, $k = 39$, $\alpha_{95} = 11°$. The diabase dips north at 4° and shows only minor alteration (epidote and chlorite). The other two sites were in a fine-grained dark layer within the sill and in a dike. No end points were observed for these sites upon alternating field and thermal demagnetization. The six diabase sites yielded a north pole at 67°E, 3°S, dm = 15, dp = 11. This pole position is similar to that obtained for the youngest volcanic suite (Flaherty Formation) of the Belcher Islands and to that obtained for red beds of the La Grande 4 outlier, suggesting a time-stratigraphic correlation between these units.

Trente-huit carottes prélevées sur huit sites localisés dans la fenêtre protérozoïque de lac Sutton que l'on croit représenter le prolongement de la ceinture Peri-Ungava. Six sites dans un filon-couche de diabase de 100 m en épaisseur ont donné essentiellement la composante unique d'aimantation avec les valeurs moyennes pour $D = 35°$, $I = -54°$, $k = 39$, $\alpha_{95} = 11°$. Le pendage de la diabase est de 4° nord et elle ne montre qu'une faible altération (épidote et chlorite). Les deux autres sites se trouvent dans une couche foncée à grain fin à l'intérieur du filon-couche et dans un dyke. Les directions finales n'ont pas été déterminées pour ces sites lors de la désaimantation par champs alternatifs et traitements thermiques. Les six sites dans la diabase ont indiqué un pôle nord à 67°E, 3°S, dm = 15, dp = 11. La position de ce pôle est semblable à celle obtenue pour la plus récente séquence volcanique (formation Flaherty) des îles Belcher et à celle obtenue pour les formations rouges de l'avant-butte de la Grande 4, suggérant une corrélation stratigraphique-temps entre ces unités.

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Introduction

The geology of the Proterozoic Sutton Lake inlier west of James Bay (Winisk) was described by Bostock (1970). His stratigraphic column shows that the rocks of interest in paleomagnetism are in the upper part of the Sutton Ranges Formation: a diabase–gabbro layer up to about 100 m thick unconformably topped by Phanerozoic sediments, and an underlying jaspelitic iron formation about 20 m thick. The main outcrop is at Sutton Narrows (54°24′N, 84°40′W) where the rocks dip 4° to the north.

Thirty-eight cores were collected at eight sites. Sites 1–7 were from Sutton Narrows. Sites 1–6 were distributed from the bottom to top of the diabase sill. Site 3 was from a fine-grained layer in the coarser diabase. Site 7 was from a dike cutting the underlying iron formation and site 8 was from massive diabase at Aquatuk Lake, 6 km southeast of Sutton Narrows. Orientation of the cores was by sun compass. Two specimens per core were used for alternating field (AF) testing and for most cores one was used for thermal testing.

Results

The natural remanent magnetization (NRM) directions generally show strong within-site scatter (Table 1) with little overall consistency. The NRM intensities are in the 10^{-2}–10^{-3} SI range (10^{-5}–10^{-6} emu cgs) and thus very low for a basic magmatic rock. A Curie temperature of 580°C indicates that the magnetic mineral is pure magnetite. Epidote and chlorite are commonly present, suggesting low-temperature alteration of mafic minerals.

The NRM was tested both by AF and thermal demagnetization. AF testing up to 100 mT (1000 Oe (79 kA m^{-1})) showed that median coercivities range from low to very high. Directional changes are generally very small in fields stronger than 30 mT but the best grouping is observed in field intensities less than 50 mT. There are no indications that the higher coercivity portion of the NRM consists of more than one component. Within-site scatter is much reduced by AF cleaning (Table 1) and there is good coherency among sites 1, 2, and 4–6. Specimens from sites 3 and 7 showed no end points. These sites were not further considered. Site 8 from Aquatuk Lake was split into 8A, two cores with a northeast-up direction, and 8B, represented by three cores suggesting a possible reversed direction or the overprint reported by Fujiwara and Schwarz (1975),

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TABLE 1. Data for sites before treatment (NRM) and after alternating field (AF) and thermal testing

	NRM						AF							THERMAL						
Sites	N	n	D (°)	I (°)	k	α_{95} (°)	N	n	Intensity (mT)	D (°)	I (°)	k	α_{95} (°)	N	n	Range (°C)	D (°)	I (°)	k	α_{95} (°)
1	5	10	286	−80	2.0	74	5	9	30–40	25	−52	285	3.1							
2	5	10	20	−44	1.5	80	5	9	30–40	32	−52	448	2.4	4	4	525–540	41	−52	106	9
3	7	14	275	−47	3.6	28														
4	3	8	48	−9	8.7	20	3	4	30–40	51	−52	225	6.1							
5	5	10	46	−35	2.0	68	4	5	30–40	35	−55	30.5	14.1	4	4	500–525	33	−66	190	7
6	5	8	65	−60	13.4	16	5	5	40–45	65	−63	19	18	5	5	500–550	35	−65	57	10
7	3	7	71	+28	19.5	14														
8A	5	10	343	+25	2.8	49	2	4	35–45	13	−41	31	17							
8B							3	4	30–45	255	+60	31	17*							
Average	Sites									35	−54	39	11				37	−61	105	12
	Specimens									33	−53	33	4				37	−61	62	5

		Pole				Pole			
		Long. (°)	Lat. (°)	k	α_{95} (°)	Long. (°)	Lat. (°)	k	α_{95} (°)
	Sites	67E	3S	15	11	68E	11S	19	14
	Specimens	68E	3S	6	4	68E	12S	8	6

NOTES: N = number of cores; n = number of specimens; D and I = declination and inclination (− is upwardly directed); k and α_{95} are commonly used statistical parameters to express grouping of directions; mT = AF peak intensity. Range in temperature employed in thermally demagnetized data is also given. Average directions and pole positions are listed giving both sites and cores unit weight. Values given after pole position are semi-axes of 95% oval of confidence in degrees of arc. All directions are given for zero bedding dip.
*Excluded in pole calculation.

Schmidt (1980), and Schwarz and Fujiwara (1981) for parts of the Circum-Ungava Belt. The average direction for sites 1, 2, 4–6, and 8A giving sites equal weight corresponds well with the average obtained when giving the specimens equal weight.

Distributed blocking temperatures (T_B) were observed for specimens from sites 3 and 7. In general, T_B's are high and, as in many of the high coercivity AF results, an initial increase in intensity is observed due to the removal of a downward viscous component during heating to 200°C. Most specimens show very little change in direction, indicating a single component magnetization with $T_B > \sim 200$°C. An adequate number of end points for statistical purposes was obtained for specimens from sites 2, 5, and 6 (Table 1). The average direction for these sites calculated on either a site or a specimen basis shows a 7–8° higher negative inclination than that obtained from the AF experiments but this difference is probably not significant in view of the grouping statistics. The mean direction for the Sutton Lake diabase (Table 1) is comparable to those determined for the Eskimo volcanics of the Belcher Islands (Schmidt 1980) and the Sakami Formation red beds of the La Grande 4 outlier east of James Bay (Schwarz and Freda 1980), indicating a time-stratigraphic relation between these units (see also Chandler and Schwarz 1980).

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Conodonts and the position of the Ordovician–Silurian boundary at the eastern end of Anticosti Island, Québec, Canada

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Conodonts have been recovered from a section spanning the Ordovician–Silurian boundary at the eastern end of Anticosti Island. They indicate that the systemic boundary lies slightly above or at the top of a pisolitic limestone bed. This placement of the boundary compares closely with that based on a previous interpretation of brachiopod faunas from the same section, and also with the position of the systemic boundary based on conodonts elsewhere on Anticosti Island. The conodonts are the first to be reported from the eastern end of Anticosti Island.

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Introduction

Anticosti Island, in the Gulf of St. Lawrence (Fig. 1), is renowned for its diversity and abundance of Late Ordovician and Early Silurian fossils. The stratigraphic sequence dips very gently to the southwest and comprises a thick, continuous sequence of mainly carbonate rocks. Deposition across the Ordovician–Silurian boundary is apparently continuous and the strata span a period of time not represented over most of the rest of North America. For these reasons Anticosti Island is currently under review as a potential stratotype for the Ordovician–Silurian boundary and for the lowest Silurian series. A considerable amount of information has recently been published documenting the strata and faunas of the island and much of this is to be found in a volume produced for a field trip undertaken in 1981 by

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